Patric O'Brien
Vice President – Regulation and Pricing

nationalgrid

March 24, 2023

VIA ELECTRONIC FILING
Honorable Michelle Phillips, Secretary
New York State Public Service Commission
3 Empire State Plaza, 19th Floor
Albany, NY 12223-1350

Re: Cases 17-E-0238 and 17-G-0239 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service

Cases 20-E-0380 and 20-G-0381 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service

Cases 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service

Cases 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service

Dear Secretary Phillips,

Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid NY, and KeySpan Gas East Corporation d/b/a National Grid (collectively, the "National Grid" or the Company") submit this petition requesting that the New York Public Service Commission approve revisions to certain Gas Business Enablement ("GBE") program key performance indicators ("KPIs"). National Grid's current rate plans authorize the Company and Department of Public Service Staff ("DPS Staff") to revise the GBE KPIs if they are no longer reasonable measures of performance. The revisions, which are supported by DPS Staff, correct errors made in the original assumptions and definitions, making the KPIs more accurate measures of performance.

Should you have any questions or concerns, please do not hesitate to contact me.

Respectfully submitted,

/s/ Patric O'Brien
Patric O'Brien

cc: Paul Darmetko (via email)
Brandon Goodrich (via email)
Aric Rider (via email)
MaryAnn Sorrentino (via email)
Allison Manz (via email)
Nicholas Forst (via email)

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION
NOTICE OF PROPOSED RULEMAKING

Pursuant to the provisions of the State Administrative Procedure Act, notice is hereby given of the following proposed rulemaking:

1. **Proposed Action:** Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid NY, and KeySpan Gas East Corporation d/b/a National Grid filed a petition pursuant to its rate plans seeking Commission approval of revisions to the GBE Program KPIs. The revisions are supported by DPS Staff and correct errors made in the original assumptions and definitions, making the KPIs more accurate measures of performance.

2. **Statutory Authority:** Public Service Law Sections 4, 5, 65, and 66.

3. **Subject of the Proposed Rule:** Revisions to GBE Program KPIs.

4. **Purpose of Proposed Rule:** Revisions to GBE Program KPIs to correct errors made in the original assumptions and definitions, making the KPIs more accurate measures of performance.

5. **Substance of the Proposed Rule:** National Grid's current rate plans authorize the Company and Department of Public Service Staff to revise the GBE KPIs if they are no longer reasonable measures of performance. The revisions, which are supported by DPS Staff, correct errors made in the original assumptions and definitions, making the KPIs more accurate measures of performance.

6. **Text of proposed rule and required statements and analyses may be obtained by filing a Document Request Form (F-96) located on the Commission's website http://www.dps.state.ny.us/f96dir.htm. For questions, contact:** Central Operations, Public Service Commission, Bldg. 3, Empire State Plaza, Albany, NY 12223-1350, (518) 474-6530.

7. **Data, views or arguments may be submitted to:** Michelle L. Phillips, Secretary, Public Service Commission, Bldg. 3, Empire State Plaza, Albany, NY 12223-1350, (518) 474-6530, email: secretary@dps.ny.gov.

8. **Public comment will be received until:** 60 days after publication of this notice.

9. **Regulatory Impact Statement, Regulatory Flexibility Analysis for Small Business and Rural Area Flexibility Analysis:** Statements and analyses are not submitted with this notice because the rule is within the definition contained in section 102(2)(a)(ii) of the State Administrative Procedure Act.

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STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Cases 17-E-0238 and 17-G-0239 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service

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Petition of
Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid NY, and KeySpan Gas East Corporation d/b/a National Grid
To Modify the Gas Business Enablement Program Key Performance Indicators

Dated: March 24, 2023

STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

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To Modify the Gas Business Enablement Program Key Performance Indicators

INTRODUCTION

Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid NY, and KeySpan Gas East Corporation d/b/a National Grid (collectively, the "Company" or "National Grid") submit this petition requesting that the New York Public Service Commission ("Commission") approve revisions to certain Gas Business Enablement ("GBE") program key performance indicators ("KPIs"). National Grid's current rate plans authorize the Company and Department of Public Service Staff ("DPS Staff") to revise the GBE KPIs if they are no longer reasonable measures of performance. The revisions, which are supported by DPS Staff, correct errors made in the original assumptions and definitions, making the KPIs more accurate measures of performance.

BACKGROUND

The GBE program is a comprehensive framework of new information technology solutions to strengthen and improve National Grid's performance. Among other things, the GBE program replaces manual, paper-driven processes with modern technical solutions. Under the terms of its rate plans, the Company is authorized to recover GBE program costs subject to certain customer protections designed to ensure the program delivers the intended benefits to customers.[1] The protections include a GBE program cost cap, downward-only tracker, and the following six KPIs measuring operational, customer, and safety performance:

i. Work Management – Customer Meter Services ("CMS KPI")
ii. Work Management – Maintenance and Construction ("M&C KPI")
iii. Engineering and Back Office
iv. Customer Call Volume
v. Customer Experience Rating ("Customer Experience KPI")
vi. Gas Safety and Compliance

The GBE KPIs were first included in the 2017 NMPC Joint Proposal[2] and then extended to KEDNY and KEDLI in the 2019 KEDNY/KEDLI Joint Proposal.[3] The KPIs are measured one-time, in the 12-month period ending March 31, 2024 ("FY24"), which is when the GBE program

[1] *See* Cases 17-E-0238 and 17-G-0239, Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service, "Order Adopting Terms of Joint Proposal, Establishing Electric and Gas Rate Plans" (issued and effective March 15, 2018) ("2017 NMPC Joint Proposal"), at Section IV.7.8; Cases 20-E-0380 and 20-G-0381, Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022) ("2020 NMPC Joint Proposal"), at Section IV.10.2; Cases 19-G-0309 and 19-G-0310, Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid for Gas Service, "Order Approving Joint Proposal, as Modified, and Imposing Additional Requirements" (issued and effective August 12, 2021) ("2019 KEDNY/KEDLI Joint Proposal"), at Section IV.6.2.

[2] 2017 NMPC Joint Proposal at Section IV.7.8.3 and Appendix 10. *See also* 2020 NMPC Joint Proposal at Section IV.10.2 and Appendix 10.

[3] 2019 KEDNY/KEDLI Joint Proposal at Section IV.6.2.3 and Appendix 10.

is expected to be in service. If the Company misses a KPI, the Company is subject to a negative revenue adjustment. If the Company overperforms on a KPI, the overperformance can be used to offset an underperforming KPI.

Typically, KPIs are set based on an historic baseline of performance supported by established ways of working. This traditional approach posed a challenge with the GBE KPIs because the system would not be in service for nearly seven years. Therefore, the Joint Proposals contained a mechanism that authorized the Company and Staff, by mutual agreement, to propose adjustments to the KPIs, if necessary, to ensure they remained a reasonable measure of performance. Specifically, the 2017 NMPC and the 2019 KEDNY/KEDLI Joint Proposals provide:

> In the event of any unforeseen circumstances, changes in law or regulation, or other events that impact the underlying assumptions used to develop the GBE KPI performance targets, such that the KPI targets are no longer reasonable measures of the Company's performance, the Company and Staff may, by mutual agreement subject to Commission approval, equitably adjust the GBE KPI targets upward or downward to address such changes.[4]

The 2017 NMPC Joint Proposal included a document entitled, "GBE Program Niagara Mohawk KPI Descriptions and Calculations," that explained the methodology to calculate the KPIs specific to NMPC ("KPI Descriptions").[5] Additionally, NMPC made a filing in the 2017 NMPC Rate Case, dated April 13, 2018, that provided additional detail on inputs to certain KPIs ("April 2018 NMPC Report") (the KPI Descriptions and April 2018 NMPC Report are collectively referred to as the "NMPC KPI Document"). To assist with monitoring the need for changes, the Joint Proposals contained annual KPI reporting requirements.

[4] 2017 NMPC Joint Proposal at Section IV.7.8.3 at 47-48; 2019 KEDNY/KEDLI Joint Proposal at Section IV.6.2.3 at 65-66; s*ee also* 2020 NMPC Joint Proposal at Section 10.2 at 55 (adopting by reference Section IV.7.8.3 of the 2017 NMPC Joint Proposal).
[5] The same document appears as Attachment 10 to the 2020 NMPC Joint Proposal at 3-6.

In preparing the report documenting KEDNY and KEDLI's GBE program KPI performance for the first time in May 2022, the Company closely examined the data and inputs, and discovered inconsistencies in how the initial baseline targets were derived for the M&C and CMS KPIs along with how the calculation methodology was described in the NMPC KPI Document for NMPC. The Company also discovered that the 2019 KEDNY/KEDLI Joint Proposal inadvertently did not include a corresponding KEDNY and KEDLI KPI Document to explain the calculations for KEDNY and KEDLI. The Company convened a meeting with DPS Staff to discuss the issues and the Company's proposed resolution. DPS Staff conducted a thorough review of the KPIs, asked discovery questions, and proposed a change to the Company's proposal.

The revisions proposed in this petition reflect the combined efforts of the Company and DPS Staff. Both parties agree that revisions are necessary to (i) the M&C KPI for NMPC, KEDNY, and KEDLI; (ii) the CMS KPI for NMPC; and (iii) minor clarifying edits to the Customer Experience KPI for NMPC, KEDNY, and KEDLI. Further, the Company and DPS Staff agree that the NMPC KPI Document should be updated, and a corresponding KEDNY and KEDLI KPI Document created, to better document the process. A redlined and a clean version of the NMPC KPI Document reflecting the updates agreed upon by the Company and DPS Staff is included in **Attachments 1 and 2**, respectively. A KEDNY and KEDLI KPI Document is included in **Attachment 3**.

The proposed KPI revisions and rationale are discussed in the sections that follow. As will be seen, the changes adhere to the intent of the Joint Proposals and ensure that the KPIs are correctly measuring the Company's performance.

DISCUSSION

1. M&C GBE KPI

A. Historic Calculation

The M&C KPI is intended to measure the average feet of gas distribution main installed per day by Company personnel. The KPI baseline and targets in the Joint Proposals are as follows:

Table 1 – Original KPI Baseline and Targets[6]

Company	KPI Baseline	FY24 KPI Target
NMPC	12.03	12.39
KEDNY	11.43	11.77
KEDLI	12.49	12.86

The NMPC KPI Document set forth the following methodology to calculate the M&C KPI for NMPC.

> The product of (i) the total number miles of gas "Distribution Main" installed by Company gas M&C FTEs in the Company's gas service territory in the fiscal year multiplied by (ii) the number of feet in a mile, divided by (iii) the number of Company gas M&C FTEs (based on average crew size during the fiscal year), with the result divided by the number of working days (240) and expressed to two decimal places.
>
> - "Distribution Main" means the following types of gas distribution main: new main, proactive, and reactive replacements, reliability and reinforcements, and city state construction replacements.
> - "Miles" means miles of gas distribution main installed.
> - "Feet" means feet of gas distribution main installed.

[6] 2017 NMPC Joint Proposal at Appendix 10, Schedule 4; 2019 KEDNY/KEDLI Joint Proposal at Appendix 10, pages 2 and 6.

- "FTEs" includes Company gas employees and excludes contractors.[7]

The 2019 KEDNY and KEDLI Joint Proposal does not contain a similar description of the M&C KPI for KEDNY and KEDLI. Therefore, to attempt to replicate the NMPC calculation methodology for purposes of reporting the KPIs for KEDNY and KEDLI, the Company closely examined the historic data and reporting. In so doing, the Company discovered inconsistencies between how the baseline was derived and the methodology that was written to calculate the M&C KPI in the NMPC KPI Document. Further, the analysis revealed that had NMPC or KEDNY and KEDLI strictly followed the NMPC KPI Document, the methodology would not have produced an accurate reflection of performance, but an anomalous result where the Company's actual performance would be well above the established targets.

In reviewing the source files for NMPC, the Company discovered three main flaws with the KPI. First, the baseline used to derive the NMPC target was based on an assumed average productivity factor using Massachusetts, not New York, data. This was likely because sufficient New York data did not exist at the time the KPI was set in 2017. Second, while the NMPC KPI Document specifically excluded contractors from the calculation, the baseline included main installation mileage associated with both contractors and in-house crews. Third, while the NMPC KPI Document references 240 working days, that variable was not used in the calculation to derive the baseline.

The Company next reviewed the source files for KEDNY and KEDLI and confirmed the same inconsistencies found with NMPC, as shown in tables 2 and 3 below.

[7] 2017 NMPC Joint Proposal at Appendix 10, Schedule 2.

Table 2 – KEDNY M&C KPI Baseline Calculation

Baseline:

values are assumed based on MA data

New York - KEDNY (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	18	miles	64	feet/day	53,033	4.5	10.56		
	Proactive and Reactive Main Replacement	21	miles	54	feet/day	75,319	4.5	10.68		
	Reliability & Reinforcements	12	miles	43	feet/day	51,592	4.5	4.69		
	CSC MRPL (includes Encroachements)	57	miles	48	feet/day	225,465	4.5	25.52		
	Total	108					4.5	51.44	11.43	11.77

Miles includes both In-house and contractors

240 working days not utilized

Table 3 – KEDLI M&C KPI Baseline Calculation

Baseline:

values are assumed based on MA data

New York - KEDLI (Original Calculatio Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	54	miles	64	feet/day	158,945	4.5	20.31		
	Proactive and Reactive Main Replacement	97	miles	54	feet/day	343,137	4.5	31.21		
	Reliability & Reinforcements	13	miles	43	feet/day	56,730	4.5	3.31		
	CSC MRPL (includes Encroachements)	5	miles	48	feet/day	18,753	4.5	1.36		
	Total	169					4.5	56.19	12.49	12.86

Miles includes both In-house and contractors

240 working days not utilized

Specifically, the Company confirmed that the baseline for the M&C KPI was derived using a three-year average of main installation from fiscal year ("FY") 16 through FY18 that included both in-house and contractor crew mileage. The three-year average was multiplied by an average productivity unit that was derived using Massachusetts assumptions. The actual miles from each main installation program were then divided by the three-year average, with the result divided by the average crew size of 4.5. This resulted in the baseline of 12.03 for NMPC, 11.43 for KEDNY, and 12.49 for KEDLI. The KPI target was derived by applying a three percent improvement factor to the baseline.

For reporting the KPIs for FY19-FY21, NMPC followed the methodology used to set the baseline, not the NMPC KPI Document. For example, the FY19-FY21 reports contained

contractor and in-house main installation mileage, did not use 240 working days, and held the average crew size constant at 4.5 instead of adjusting it each FY. The reports also used the average productivity unit that was based on Massachusetts data. Importantly, had NMPC followed the NMPC KPI Document, and were KEDNY and KEDLI to do the same, actual FY22 M&C KPI performance for all companies would be far above the target as shown in table 4 below.

Table 4 – Actual Results if the NMPC Document were Followed

Company	Target	Actuals Following NMPC KPI Document
NMPC	12.39 ft	323 ft
KEDNY	11.77 ft	377 ft
KEDLI	12.86 ft	863 ft

B. Revised Calculations for NMPC, KEDNY, and KEDLI

In light of the issues identified above, the Company and DPS Staff considered the following three options before mutually agreeing that a revised calculation best met the intent of the GBE KPIs and the interest of customers.

- ***Leave Calculation as is and do Nothing.*** This option was rejected because the historic calculation utilized by NMPC does not align with the NMPC KPI Document. The historic calculation includes both in-house and contractor crews, does not adjust crew size, and does not use 240 working days. In addition, the historic calculation is based on an average feet per day assumption that came from Massachusetts data; therefore, it is not a reasonable measure of New York performance. Further, the NMPC KPI Document methodology could not be applied to KEDNY and KEDLI because those companies primarily use contractor crews (as does NMPC) – and that data was included in the baseline.

- ***Calculate per the NMPC KPI Document.*** This option was similarly rejected because NMPC, KEDNY, and KEDLI's actual performance would so greatly exceed the targets that the KPI would effectively be rendered meaningless. The result would not be consistent with the intent of the GBE KPIs or in the best interest of customers.
- ***Revise the Existing Calculation.*** This is the recommended option. The revised calculation is included in Attachments 1 and 2 for NMPC and Attachment 3 for KEDNY and KEDLI.

There are many benefits to the revised calculation. The revised calculation holds true to the original baseline calculation with an adjustment to the productivity factor to make it New York specific – and a true measurement of performance. The methodology aligns with how the Company tracks M&C distribution main work (and will track going forward with the GBE program), reflecting improvements in how data is tracked since the KPI was originally developed in 2017. Including contractors in the calculation is reasonable and an improvement as the change aligns with the baseline, and the GBE program will benefit both in-house and contractor crews. Indeed, the majority of M&C distribution main work is performed by contractor crews, especially in KEDNY and KEDLI's service territory. Further, the revisions enable the Company and DPS Staff to create a KEDNY and KEDLI KPI Document and update the NMPC KPI Document. The result is an improved process – one that supports an effective audit trail in the future. DPS Staff and the Company also added as an appendix to both the NMPC and KEDNY and KEDLI KPI Documents a spreadsheet that details the calculation, further enhancing transparency. Lastly, the revisions provide a consistent methodology across the three companies. For all these reasons, the revisions to the M&C GBE KPI result in a more accurate measure of performance.

To ensure no change in the operation of the mechanism and that the parameters of the baseline and target remain consistent with the Joint Proposal, the Company proposed using FY22 actuals under the new methodology as the new NMPC, KEDNY, and KEDLI baselines. The NMPC, KEDNY, and KEDLI targets would be set by applying the differential between the original baseline and target.

DPS Staff agreed with using FY22 actuals as the new baselines but believed that the targets should be set by applying the original three percent improvement factor to the new baselines. While this would result in the Company having to deliver three percent higher performance in a shorter period than originally contemplated in the Joint Proposals, the Company agrees with DPS Staff's position.

Table 5 shows the revised baseline and targets for NMPC, KEDNY, and KEDLI for the M&C KPI.

Table 5 – M&C KPI New Baseline and Targets

Company	KPI Baseline (FY22)	FY24 KPI Target
NMPC	39.13	40.30
KEDNY	9.46	9.74
KEDLI	36.37	37.46

2. CMS GBE KPI

A. Historic Calculation

The CMS KPI is intended to measure the average number of gas jobs completed per day by Company personnel. The KPI baseline and target in the 2017 NMPC Joint Proposal is as follows:

Table 6 – Original KPI Baseline and Targets[8]

Company	KPI Baseline	FY24 KPI Target
NMPC	2.7	2.78

The NMPC KPI Document set forth the following methodology to calculate the CMS KPI for NMPC.

> The quotient of (i) the total number of Jobs completed by CMS FTEs during the fiscal year, divided by (ii) the total number of gas CMS FTEs during the fiscal year, with the result divided by the number of working days (240) and expressed to two decimal places.
>
> - "FTEs" includes Company gas employees and excludes contractors.
> - "Jobs" means all Company CMS planned and collections jobs completed by CMS FTEs.

The April 2018 NMPC Report included a listing of the "Jobs" included in the KPI.[9] As with the M&C KPI, while the NMPC KPI Document set forth the process to calculate the CMS KPI for NMPC, the historic calculation utilized in the baseline does not align with this methodology.

Specifically, total gas CMS jobs were utilized in one part of the calculation; however, CMS FTEs included both electric and gas FTEs in the other portion of the calculation. The use of both electric and gas FTEs applied to gas CMS jobs seems inconsistent from a metric perspective and certainly inconsistent with the algorithm set forth in the NMPC KPI Document. Notwithstanding, that was how the baseline was derived. For Collections Jobs, NMPC applied a 14 percent assumption to total Collections Jobs Completed to represent the gas portion of those jobs. This

[8] 2017 NMPC Joint Proposal at Appendix 10, Schedule 4.
[9] April 2018 NMPC Report at 2.

collections assumption is not addressed in the NMPC KPI Document and was held constant although the actual percentage varied year-to-year.

B. Revised Calculation for NMPC

In light of these issues, the Company and DPS Staff considered the following three options:

- ***Leave Calculation as is and do Nothing.*** This option was rejected because the historic calculation utilized by NMPC does not comport with the NMPC KPI Document. The historic calculation included gas CMS jobs in the numerator but gas and electric CMS FTEs in the denominator, which is inconsistent with the NMPC KPI Document.
- ***Calculate per the NMPC KPI Document.*** This option was similarly rejected because it would change how the baseline was derived, resulting in NMPC's actual performance greatly exceeding the target (*i.e.,* NMPC's FY22 performance would be 8.02). Such a result would render the KPI effectively meaningless.
- ***Revise the Existing Calculation.*** This is the recommended option. The revision adds electric and gas CMS jobs to the numerator, aligning it with the denominator, which already included electric and gas CMS FTEs. This results in a like for like measurement. The revised calculation is included in Attachments 1 and 2 for NMPC.

The benefits of the Company and DPS Staff's revision is that it is a simple fix that brings consistency to the calculation. It also eliminates the need to include a collections assumption as now all jobs are included in the metric. Inasmuch as the GBE program benefits both electric and gas CMS employees in NMPC's service territory, the change aligns with the intent of the KPI, which is to measure NMPC's actual performance. The revision updates the "Jobs" as those listed in the April 2018 NMPC Report were pre-GBE and have changed since the GBE program was implemented. While no change is required for KEDNY and KEDLI, there is a need for a KEDNY

and KEDLI KPI Document that defines the calculation and inputs. The KEDNY and KEDLI KPI Document is included in Attachment 2, bringing enhanced transparency to the calculation.

Similar to the M&C KPI, the Company initially proposed using FY22 actuals under the new methodology as the new NMPC baseline. The NMPC target would be set by applying the differential between the original baseline and target to the new baseline. DPS Staff agreed with using FY22 actuals as the new baseline but proposed that the target be set by applying the original three percent improvement factor to the new baseline. The Company agrees with DPS Staff's position for the same reasons discussed above. Table 7 shows the revised baseline and target for NMPC.

Table 7 – CMS KPI New Baseline and Target

Company	KPI Baseline (FY22)	FY24 KPI Target
NMPC	6.31	6.50

3. Customer Experience KPI

The Customer Experience KPI measures the overall satisfaction of the Company's residential customers captured in the "Ease of Doing Business" survey. The Company and Staff propose two minor revisions to the NMPC KPI Document.

The first revision reflects that all residential customers, both electric and gas, are included in the "Ease of Doing Business" survey. The NMPC KPI Document inadvertently mentioned "residential gas customers" but the survey includes both. The second revision simply clarifies that there is only one "Ease of Doing Business" survey question, not "questions," as incorrectly stated in the NMPC KPI Document. The question utilized for the KPI is "How easy is it to do business with National Grid." Attachments 1 and 2 contain the changes for NMPC. The new KEDNY and

KEDLI KPI Document, which includes the same definition as NMPC, is included in Attachment 3.

CONCLUSION

The 2017 NMPC Joint Proposal and the 2019 KEDNY and KEDLI Joint Proposals recognized the likelihood that changes would be needed to the GBE program KPIs given the time between when the metrics were set and GBE implementation. The changes mutually agreed upon by the Company and DPS Staff correct inconsistencies due to enhancements in assumptions and data collection that, while reasonable at the time the KPIs were adopted, are no longer reasonable. Importantly, the changes ensure that the KPIs correctly measure the Company's performance in a manner that fully adheres with the intent of the Joint Proposals. The changes provide the added benefit of having documentation in place that conforms with how the KPIs are actually calculated, enabling a path forward to support any future reviews of the metrics. For all these reasons, the Company and DPS Staff respectfully request that the Commission approve the instant petition.

Respectfully submitted,

THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY, KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID, AND NIAGARA MOHAWK POWER CORPORATION D/B/A NATIONAL GRID

By: *Patric O'Brien*

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Dated: March 24, 2023

Attachment 1

Redlined NMPC KPI Document

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr><th>Measurement Area</th><th>KPI</th><th>Metric Description</th><th>Algorithm</th><th>Data Source</th></tr>
<tr><td>Work Management-Customer Meter Services ("CMS")</td><td>Avg. # of completed Jobs per CMS FTE (jobs/day)</td><td>Measures the average number of jobs completed per day by each working CMS FTE</td><td>The quotient of (i) the total number of Jobs completed by CMS gas and electric FTEs during the fiscal year, divided by (ii) the total number of gas and electric CMS FTEs during the fiscal year, with the result divided by the number of working days (240) and expressed to two decimal places.

CMS KPI: A/B/C
<table><tr><th>Variables</th><th>Description</th></tr><tr><td>A</td><td>Total CMS Jobs Completed including Collections</td></tr><tr><td>B</td><td>Total CMS FTE Count</td></tr><tr><td>C</td><td>Working Days in a Year</td></tr></table>"FTEs" includes Company gas and electric employees and excludes contractors

"Jobs" means all Company CMS planned and collections jobs completed by CMS FTEs

"Jobs" utilized to calculate the KPI baseline and the actual KPI performance are listed below:

CMS Planned Jobs:
Emergency Electric & Gas - O&M
Turn-On Turn-Off – O&M
Investigate and Read Meter – O&M
Inspection-Asset – O&M
Inspection Location Total – O&M
Meter Asset Work – O&M</td><td>Jobs Data and HRIS system</td></tr>
</table>

Deleted: gas

Commented [OPRS1]: There were three changes made:
1. Added electric.
2. Added a table showing the formula just for clarity purposes.
3. The "Jobs" input was not in Appendix 10 but the separate August 2018 filing. I added them here for clarity. Note that the August 2018 filing included pre-GBE "Jobs" naming conventions. Post-GBE the naming conventions have changed. The new names are set forth here. While the names have changed, the "Jobs" are the same.

Deleted: gas

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr><td></td><td></td><td></td><td>Leak Surveillance, Recheck and Inspection – O&M
Leak Monitoring – O&M
Meter Asset Work – Capital
Maintain Service – Capital

Collections Jobs:
Completed Visits – Terminations and Collections</td><td></td></tr>
<tr><td>Work Management-Maintenance & Construction ("M&C")</td><td>Avg. ft. of gas distribution main installed or replaced per M&C and contractor FTE (ft/day)</td><td>Measures the average feet of gas distribution main installed per day by each working gas M&C and contractor FTE</td><td>The formula to calculate the KPI is:

M&C KPI $= \frac{J}{K}$ where

J = ((A*E/I) + (B*F/I) + (C*G/I) + (D*H/I))

K = Total number of gas in-house / contractor FTEs (based on average crew size during the fiscal year)

<table>
<tr><th>Variables</th><th>Description</th></tr>
<tr><td>A</td><td>Growth Main Miles</td></tr>
<tr><td>B</td><td>Proactive and Reactive Main Replacement Miles</td></tr>
<tr><td>C</td><td>Reliability & Reinforcements Miles</td></tr>
<tr><td>D</td><td>CSC MRPL (includes Encroachments) Miles</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>E</td><td>Avg. Feet Per Day - Growth Main</td></tr>
<tr><td>F</td><td>Avg. Feet Per Day - Proactive and Reactive Main Replacement</td></tr>
</table>
</td><td>M&C Workplan</td></tr>
</table>

Deleted: The product of (i) the total number miles of gas "Distribution Main" installed by Company gas M&C FTEs in the Company's gas service territory in the fiscal year *multiplied by* (ii) the number of feet in a mile, *divided by* (iii) number of Company gas M&C FTEs (based on average crew size during the fiscal year), with the result divided by the number of working days (240) and expressed to two decimal places.¶
"Distribution Main" means the following types of gas distribution main: new main, proactive and reactive replacements, reliability and reinforcements, and city state construction replacements ¶
"Miles" means miles of gas distribution main installed ¶
"Feet" means feet of gas distribution main installed¶
¶
"FTEs" includes Company gas employees and excludes contractors ¶
¶
The average crew size used in calculating the KPI baseline is 4.5 for the following types of M&C work: Growth Main,¶
Proactive and Reactive Main Replacement,¶
Reliability & Reinforcements, City State¶
Construction Main Replacement (includes¶
Encroachments). Because crew size varies day-by-day, the average crew size represents the average for all National Grid operating companies for the M&C Main Programs listed above.¶
¶
For purposes of calculating actual KPI performance for each Rate Year as required by Section 7.8.4 of the JP and for FY23 under Section 7.8.3 of the JP, the¶
Company will calculate the average crew size for the given FY for M&C Main Programs referenced above.¶

Commented [OPRS2]: The change was to update the inputs as discussed during the September 14, 2022 meeting.

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>
<table>
<tr><td>G</td><td>Avg. Feet Per Day - Reliability & Reinforcements</td></tr>
<tr><td>H</td><td>Avg. Feet Per Day - CSC MRPL (includes Encroachments)</td></tr>
<tr><td>J</td><td>Jurisdiction Average (Mains)</td></tr>
<tr><td>K</td><td>The number of gas in-house / contractors FTEs (based on average crew size during the Fiscal Year)</td></tr>
</table>

Variables E, F, G, and H are calculated as follows for in-house:
<table>
<tr><td>EFGH</td><td>Average Feet Per Day (In-House)</td></tr>
<tr><td></td><td>EFGH = (I*L)/P*Q*O</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>L</td><td>5280 (# of feet in a mile)</td></tr>
<tr><td>O</td><td>FTEs per Crew (varies by region)</td></tr>
<tr><td>P</td><td>Total Hours Worked</td></tr>
<tr><td>Q</td><td>8 hours daily</td></tr>
</table>

Variables E, F, G, and H are calculated as follows for contractors:
<table>
<tr><td>EFGH</td><td>Average Feet Per Day (Contractor)</td></tr>
<tr><td></td><td>EFGH = R*O</td></tr>
<tr><td>R</td><td>Average benchmark breakdown from Gas Construction (Fixed Annually)</td></tr>
<tr><td>O</td><td>FTEs per Crew (varies by region) (Fixed Annually)</td></tr>
</table>

An Excel file illustrating the calculation is included as Appendix 10a.
</td>
<td></td>
</tr>
<tr>
<td>Engineering and Back Office</td>
<td>Work Orders Processed/ Back Office FTEs (WOs/yr)</td>
<td>Measures the average number of Work Orders processed per year</td>
<td>The quotient of (i) Company total Work Orders in STORMS during the fiscal year divided by (ii) average number of Clerks active during the fiscal year.

"Work Orders" means work orders marked as closed in STORMS</td>
<td>STORMS Data</td>
</tr>
</table>

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

		in the back office by each engineering clerical employee	"Clerks" means those Company and Service Company employee and contractor roles responsible for processing Work Orders. The number of Clerks responsible for processing Work Orders utilized in calculating the KPI baseline is 26 comprising of: Administrative Clerk A (Niagara Mohawk employees – 6) Junior Clerk (Niagara Mohawk employee – 1) Contractors (Niagara Mohawk – 9, National Grid USA Service Company, Inc. – 10) For purposes of calculating actual KPI performance for each Rate Year as required by Section 7.8.4 of the JP and for FY23 under Section 7.8.3 of the JP, the Company will calculate the number of Clerks for the given FY based on the roles listed above for Niagara Mohawk and National Grid USA, Inc. Service Company.	R850 completion Clerical information (HRIS) and filtered for those responsible for processing work orders
Customer	Total Call Volume (Calls/year)	Measures the total volume of move and non-move telephone calls received by the Company's call centers from customers	The sum of (i) the total number of Move Calls received by the call centers from Company's customers during the fiscal year *plus* (ii) the total number of field-related Non-move Calls received by the call centers from Company's customers during the same measurement window. "Move Calls" means calls for service installation or discontinuance. "Non-move Calls" means calls to resolve a customer issue.	Move Call Benefits – call center query for fiscal year Non Move Call Benefits – call center query for fiscal year Move and Non-move Calls are included but not

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

				differentiated in the total Annual Call Volumes reported in the annual Service Quality Assurance Program report (discussed in Appendix 15) provided on a calendar year basis.
Customer / Key Measurement	Customer Experience (Effort) Rating	Measures the overall satisfaction of National Grid residential customers captured in the "Ease of Doing Business" survey, the results of which are mainly driven by number handoffs between Company departments and ease of issue resolutions	Average of scores for the "Ease of Doing Business" survey question to Company customers during the fiscal year, with the result expressed to two decimal places.	Ease of Doing Business Results from the Brand, Image & Relationship Survey

Deleted: Customer Effort Survey

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Commented [OPRS3]: The changes are:
1.Reflect that the survey includes both electric and gas customers.
2.Correct the name of the survey.

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GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

Gas Safety & Compliance	Total non-compliance occurrences	Measures the total number of gas safety non-compliance occurrences	Total occurrences of violations of the Commission's gas safety regulations, as measured by the final results of Staff's annual records and field audits.	NMPC Records and Field Audit Results Cumulative Annual Non-Compliance Occurrences (as discussed in Joint Proposal Section 12.5) from Pipeline Safety & Compliance Team

Deleted: Notes: ¶
The purpose of each KPI is to measure the performance of the Company or its employees.¶
Within 30 days of a Commission order approving the Joint Proposal in this proceeding, the Company will file with the Secretary a report containing (i) the Work Management - Customer Meter Services "Jobs," defined as all CMS planned and collections jobs completed by Company CMS FTEs; (ii) the current Work Management - Maintenance & Construction average crew size; and (iii) the current number of Engineering and Back Office "Clerks," defined as Company and Service Company employees and contractors responsible for processing Work Orders.¶

Baseline:

values are assumed based on MA data

New York - NiMo (New Calculation using footage)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	34	miles	62	feet/day	103,133	4.5	21.16		
	Proactive and Reactive Main Replacement	45	miles	51	feet/day	167,619	4.5	28.29		
	Reliability & Reinforcements	2	miles	43	feet/day	10,669	4.5	1.50		
	CSC MRPL (includes Encroachements)	5	miles	47	feet/day	20,809	4.5	3.20		
	Total	**86**					**4.5**	**54.14**	**12.03**	**12.39**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	39	33	29	feet/day/crew	34
Proactive and Reactive Main Replacement	35	43	55	feet/day/crew	44
Proactive and Reactive Main Replacement	0	1	1	feet/day/crew	1
CSC MRPL (includes Encroachements)	6	4	5	feet/day/crew	5
CSC MRPL (includes Encroachements)	0	0	0	feet/day/crew	0
Reliability & Reinforcements	1	4	2	feet/day/crew	2
Total	**81**	**85**	**92**		86

NiMo Current Calculation

FY18

Growth Main	28	miles	62	feet/day	85,181	4.5	16.82		
Proactive and Reactive Main Replacement	51	miles	51	feet/day	189,116	4.5	30.72		
Reliability & Reinforcements	3	miles	43	feet/day	15,332	4.5	2.08		
CSC MRPL (includes Encroachements)	6	miles	47	feet/day	24,297	4.5	3.60		Feet per FTE
Total	**88**					**4.5**	**53.22**		**11.83** FY18

FY19

Growth Main	18	miles	62	feet/day	56,598	4.5	12.94		
Proactive and Reactive Main Replacement	50	miles	51	feet/day	186,109	4.5	35.00		
Reliability & Reinforcements	5	miles	43	feet/day	22,695	4.5	3.56		
CSC MRPL (includes Encroachements)	1	miles	47	feet/day	5,791	4.5	0.99		Feet per FTE
Total	**75**					**4.5**	**52.49**		**11.66** FY19

FY20

Growth Main	24	miles	62	feet/day	73,801	4.5	15.82		
Proactive and Reactive Main Replacement	46	miles	51	feet/day	170,796	4.5	30.11		
Reliability & Reinforcements	7	miles	43	feet/day	32,194	4.5	4.73		
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	12,467	4.5	2.00		Feet per FTE
Total	**80**				**289,258**	**4.5**	**52.67**		**11.70** FY20

FY21

Growth Main	17	miles	62	feet/day	50,770	4.5	12.44		
Proactive and Reactive Main Replacement	48	miles	51	feet/day	179,219	4.5	36.13		
Reliability & Reinforcements	4	miles	43	feet/day	16,139	4.5	2.71		
CSC MRPL (includes Encroachements)	2	miles	47	feet/day	6,860	4.5	1.26		Feet per FTE
Total	**70**				**252,988**	**4.5**	**52.54**		**11.68** FY21

FY22

Growth Main	17	miles	62	feet/day	53,328	4.5	13.96		
Proactive and Reactive Main Replacement	46	miles	51	feet/day	169,710	4.5	36.55		
Reliability & Reinforcements	1	miles	43	feet/day	3,381	4.5	0.61		
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	10,377	4.5	2.04		Feet per FTE
Total	**66**				**236,797**	**4.5**	**53.16**		**11.81** FY22

324

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 17	miles	(E) 169	feet/day	16,292	3.8	44.42		
Proactive and Reactive Main Replacement	(B) 46	miles	(F) 139	feet/day	51,789	3.8	95.76		
Reliability & Reinforcements	(C) 1	miles	(G) 154	feet/day	777	3.8	1.76		
CSC MRPL (includes Encroachements)	(D) 3	miles	(H) 125	feet/day	3,211	3.8	4.80		Feet per FTE
Total	**(I) 66**				**72,069**	**(K) 3.8**	**(J) 146.74**		**39.13** FY22

Baseline:

values are assumed based on MA data

New York - KEDNY (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	18	miles	64	feet/day	53,033	4.5	10.56		
	Proactive and Reactive Main Replacement	21	miles	54	feet/day	75,319	4.5	10.68		
	Reliability & Reinforcements	12	miles	43	feet/day	51,592	4.5	4.69		
	CSC MRPL (includes Encroachements)	57	miles	48	feet/day	225,465	4.5	25.52		
	Total	**108**					**4.5**	**51.44**	**11.43**	**11.77**

Data FY16 - FY17	Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
	Growth Main	18	16	20	feet/day/crew	18
	Proactive and Reactive Main Replacement	17	18	29	feet/day/crew	21
	Reliability & Reinforcements	14	10	11	feet/day/crew	12
	CSC MRPL (includes Encroachements)	38	67	67	feet/day/crew	57
	Total	**88**	**111**	**126**		**108**

(Original Baseline Calculation)

FY18										
	Growth Main	9	miles	64	feet/day	28,019	4.5	5.05		
	Proactive and Reactive Main Replacement	31	miles	54	feet/day	108,581	4.5	13.92		
	Reliability & Reinforcements	8	miles	43	feet/day	33,496	4.5	2.76		
	CSC MRPL (includes Encroachements)	72	miles	48	feet/day	282,578	4.5	28.93		Feet per FTE
	Total	**120**					**4.5**	**50.66**		**11.26** FY18

FY19										
	Growth Main	7	miles	64	feet/day	22,184	4.5	4.46		
	Proactive and Reactive Main Replacement	35	miles	54	feet/day	121,515	4.5	17.39		
	Reliability & Reinforcements	7	miles	43	feet/day	32,152	4.5	2.95		
	CSC MRPL (includes Encroachements)	58	miles	48	feet/day	228,147	4.5	26.06		Feet per FTE
	Total	**107**					**4.5**	**50.86**		**11.30** FY19

FY20										
	Growth Main	5	miles	64	feet/day	14,504	4.5	3.16		
	Proactive and Reactive Main Replacement	41	miles	54	feet/day	143,457	4.5	22.22		
	Reliability & Reinforcements	9	miles	43	feet/day	38,451	4.5	3.82		
	CSC MRPL (includes Encroachements)	45	miles	48	feet/day	175,955	4.5	21.76		Feet per FTE
	Total	**99**				**372,367**	**4.5**	**50.95**		**11.32** FY20

FY21										
	Growth Main	3	miles	64	feet/day	9,803	4.5	3.28		
	Proactive and Reactive Main Replacement	18	miles	54	feet/day	63,055	4.5	15.03		
	Reliability & Reinforcements	4	miles	43	feet/day	19,355	4.5	2.96		
	CSC MRPL (includes Encroachements)	39	miles	48	feet/day	152,651	4.5	29.05		Feet per FTE
	Total	**64**				**244,864**	**4.5**	**50.32**		**11.18** FY21

FY22										
	Growth Main	2	miles	64	feet/day	6,914	4.5	1.93		
	Proactive and Reactive Main Replacement	29	miles	54	feet/day	102,867	4.5	20.46		
	Reliability & Reinforcements	4	miles	43	feet/day	17,237	4.5	2.20		
	CSC MRPL (includes Encroachements)	42	miles	48	feet/day	164,038	4.5	26.05		Feet per FTE
	Total	**77**				**291,055**	**4.5**	**50.65**		**11.26** FY22

377

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22										
	Growth Main	(A) 2	miles	(E) 18	feet/day	27,816	5.0	0.53		
	Proactive and Reactive Main Replacement	(B) 29	miles	(F) 47	feet/day	132,369	5.0	17.67		
	Reliability & Reinforcements	(C) 4	miles	(G) 37	feet/day	22,242	5.0	1.89		
	CSC MRPL (includes Encroachements)	(D) 42	miles	(H) 50	feet/day	174,600	5.0	27.20		Feet per FTE
	Total	**(I) 77**				**357,027**	**(K) 5.0**	**(J) 47.30**		**9.46** FY22

Baseline:

values are assumed based on MA data

New York - KEDLI (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	54	miles	64	feet/day	158,945	4.5	20.31		
	Proactive and Reactive Main Replacement	97	miles	54	feet/day	343,137	4.5	31.21		
	Reliability & Reinforcements	13	miles	43	feet/day	56,730	4.5	3.31		
	CSC MRPL (includes Encroachements)	5	miles	48	feet/day	18,753	4.5	1.36		
	Total	**169**					**4.5**	**56.19**	**12.49**	**12.86**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	59	47	55	feet/day/crew	54
Proactive and Reactive Main Replacement	79	96	118	feet/day/crew	97
Reliability & Reinforcements	15	14	10	feet/day/crew	13
CSC MRPL (includes Encroachements)	4	3	7	feet/day/crew	5
Total	**156**	**160**	**190**		**169**

(Original Baseline Calculation)

FY18

Growth Main	51	miles	64	feet/day	150,787	4.5	17.76	
Proactive and Reactive Main Replacement	117	miles	54	feet/day	412,182	4.5	34.55	
Reliability & Reinforcements	13	miles	43	feet/day	58,074	4.5	3.12	
CSC MRPL (includes Encroachements)	2	miles	48	feet/day	7,561	4.5	0.51	
Total	**183**					**4.5**	**55.94**	

Feet per FTE **12.43** FY18

Rounded:

FY19

Growth Main	58	miles	64	feet/day	172,130	4.5	16.30	
Proactive and Reactive Main Replacement	160	miles	54	feet/day	564,589	4.5	38.07	
Reliability & Reinforcements	8	miles	43	feet/day	37,284	4.5	1.61	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	2,622	4.5	0.14	
Total	**227**					**4.5**	**56.13**	

Feet per FTE **12.47** FY19

FY20

Growth Main	31	miles	64	feet/day	91,444	4.5	9.63	
Proactive and Reactive Main Replacement	166	miles	54	feet/day	585,336	4.5	43.90	
Reliability & Reinforcements	7	miles	43	feet/day	31,355	4.5	1.51	
CSC MRPL (includes Encroachements)	0	miles	48	feet/day	1,229	4.5	0.07	
Total	**205**				**709,363**	**4.5**	**55.12**	

Feet per FTE **12.25** FY20

FY21

Growth Main	23	miles	64	feet/day	69,755	4.5	11.25	
Proactive and Reactive Main Replacement	103	miles	54	feet/day	363,501	4.5	41.75	
Reliability & Reinforcements	6	miles	43	feet/day	25,212	4.5	1.86	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	4,277	4.5	0.39	
Total	**134**				**462,745**	**4.5**	**55.25**	

Feet per FTE **12.28** FY21

FY22

Growth Main	32	miles	64	feet/day	94,586	4.5	11.54	
Proactive and Reactive Main Replacement	137	miles	54	feet/day	482,638	4.5	41.94	
Reliability & Reinforcements	7	miles	43	feet/day	28,992	4.5	1.62	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	3,951	4.5	0.27	
Total	**177**				**610,167**	**4.5**	**55.37**	

Feet per FTE **12.30** FY22

863

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 32	miles	(E) 105	feet/day	54,576	4.3	18.90	
Proactive and Reactive Main Replacement	(B) 137	miles	(F) 167	feet/day	147,038	4.3	130.00	
Reliability & Reinforcements	(C) 7	miles	(G) 148	feet/day	8,022	4.3	5.52	
CSC MRPL (includes Encroachements)	(D) 1	miles	(H) 26	feet/day	7,013	4.3	0.15	
Total	**(I) 177**				**216,649**	**(K) 4.3**	**(J) 154.56**	

Feet per FTE **36.37** FY22

KEDNY

Contractor	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	8,245	5.0	5.1	220.0	7.30	36.5
Proactive and Reactive Main Replacement	25	133,560	5.0	60.9	220.0	9.97	49.8
Reliability & Reinforcements	4	20,589	5.0	11.8	220.0	7.94	39.7
CSC MRPL (includes Encroachements)	34	178,734	5.0	65.8	220.0	12.35	61.8
Total	**65**	**341,128**	**5.0**	**143.6**	**220.0**	**10.80**	**53.99**

In-House	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	1	4,046	5.0	12.4	176.0	1.85	9.2
Proactive and Reactive Main Replacement	4	20,740	5.0	22.6	176.0	5.20	26.0
Reliability & Reinforcements	0	119	5.0	2.3	176.0	0.30	1.5
CSC MRPL (includes Encroachements)	8	41,123	5.0	44.5	176.0	5.25	26.3
Total	**13**	**66,028**	**5.0**	**81.8**	**176.0**	**4.59**	**22.93**

ALL	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	12,291	5.0	17.6	198.0	3.53	18
Proactive and Reactive Main Replacement	29	154,300	5.0	83.6	198.0	9.33	47
Reliability & Reinforcements	4	20,708	5.0	14.0	198.0	7.45	37
CSC MRPL (includes Encroachements)	42	219,857	5.0	110.2	198.0	10.07	50
Total	**(I) 77**	**407,156**	**(O) 5.0**	**(M) 225.4**	**(N) 198.0**		**(EFGH)**

KEDLI

Contractor	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	86,011	3.3	5.0	16.7	220	23.48	117.39
Proactive and Reactive Main Replacement	104	546,775	10.9	5.0	54.4	220	45.71	228.57
Reliability & Reinforcements	5	26,097	0.8	5.0	3.9	220	30.61	153.06
CSC MRPL (includes Encroachements)	1	5,296	0.9	5.0	4.4	220	5.50	27.51
Total	**126**	**664,179**	**15.9**	**5.0**	**79.3**	**220**	**38.08**	**190.42**

In-House	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	82,141	4.8	3.5	16.7	176	27.92	97.71
Proactive and Reactive Main Replacement	34	177,182	11.0	3.5	38.4	176	26.22	91.79
Reliability & Reinforcements	2	8,734	0.4	3.5	1.5	176	34.03	119.10
CSC MRPL (includes Encroachements)	-	-	0.2	3.5	0.6	176	-	-
Total	**51**	**268,057**	**16.3**	**3.5**	**57.1**	**176**	**26.65**	**93.28**

ALL	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	32	168,152	8.1	4.3	34.5	198.0	24.65	104.76
Proactive and Reactive Main Replacement	137	723,957	21.8	4.3	92.8	198.0	39.39	167.40
Reliability & Reinforcements	7	34,831	1.2	4.3	5.1	198.0	34.73	147.62
CSC MRPL (includes Encroachements)	1	5,296	1.0	4.3	4.4	198.0	6.04	25.68
Total	**(I) 177**	**932,236**	**32.2**	**(O) 4.3**	**(M) 136.8**	**(N) 198.0**		**(EFGH)**

NiMo

Contractor	FY22 (Miles)	FY22 (Feet)	Hours (Not Avail.)	Feet Per Hour (Not Avail.)	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	4	18,558	-	-	4.0	100.00	400.0
Replacement Main	29	153,259	-	-	4.0	50.00	200.0
DOT	2	8,055	-	-	4.0	50.00	200.0
Encroachments	-	-	-	-	4.0	-	-
Uprate	-	-	-	-	4.0	-	-
Total	**34**	**179,872**		**-**	**(O) 4.0**	**(R)**	**(EFGH)**

In-House	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	14	73,284	18,795	3.9	3.5	31.19	109.2
Replacement Main	17	87,164	48,004	1.8	3.5	14.53	50.8
DOT	1	5,349	5,048	1.1	3.5	8.48	29.7
Encroachments	0	888	5,519	0.2	3.5	1.29	4.5
Uprate	1	3,104	939	3.3	3.5	26.45	92.6
Total	**(I) 32**	**169,789**	**(P) 78,304**	**2.2**	**(O) 3.5**		**(EFGH)**

ALL	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	17	91,842	-	-	3.75	45.1	169.1
Replacement Main	46	240,423	-	-	3.75	37.1	139.3
DOT	3	13,404	-	-	3.75	33.4	125.2
Encroachments	0	888	-	-	3.75	1.3	4.8
Uprate	1	3,104	-	-	3.75	26.4	99.2
Total	**66**	**349,661**		**-**	**3.75**	**-**	

Attachment 2

Clean NMPC KPI Document

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr><th>Measurement Area</th><th>KPI</th><th>Metric Description</th><th>Algorithm</th><th>Data Source</th></tr>
<tr><td>Work Management-Customer Meter Services ("CMS")</td><td>Avg. # of completed Jobs per CMS FTE (jobs/day)</td><td>Measures the average number of jobs completed per day by each working CMS FTE</td><td>The quotient of (i) the total number of Jobs completed by CMS gas and electric FTEs during the fiscal year, divided by (ii) the total number of gas and electric CMS FTEs during the fiscal year, with the result divided by the number of working days (240) and expressed to two decimal places.

CMS KPI: A/B/C
<table><tr><th>Variables</th><th>Description</th></tr><tr><td>A</td><td>Total CMS Jobs Completed including Collections</td></tr><tr><td>B</td><td>Total CMS FTE Count</td></tr><tr><td>C</td><td>Working Days in a Year</td></tr></table>"FTEs" includes Company gas and electric employees and excludes contractors

"Jobs" means all Company CMS planned and collections jobs completed by CMS FTEs

"Jobs" utilized to calculate the KPI baseline and the actual KPI performance are listed below:

CMS Planned Jobs:
Emergency Electric & Gas - O&M
Turn-On Turn-Off – O&M
Investigate and Read Meter – O&M
Inspection-Asset – O&M
Inspection Location Total – O&M
Meter Asset Work – O&M</td><td>Jobs Data and HRIS system</td></tr>
</table>

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>Leak Surveillance, Recheck and Inspection – O&M
Leak Monitoring – O&M
Meter Asset Work – Capital
Maintain Service – Capital

Collections Jobs:
Completed Visits – Terminations and Collections</td>
<td></td>
</tr>
<tr>
<td>Work Management-Maintenance & Construction ("M&C")</td>
<td>Avg. ft. of gas distribution main installed or replaced per M&C and contractor FTE (ft/day)</td>
<td>Measures the average feet of gas distribution main installed per day by each working gas M&C and contractor FTE</td>
<td>The formula to calculate the KPI is:

M&C KPI $= \frac{J}{K}$ where

J = ((A*E/I) + (B*F/I) + (C*G/I) + (D*H/I))

K = Total number of gas in-house / contractor FTEs (based on average crew size during the fiscal year)

<table>
<tr><th>Variables</th><th>Description</th></tr>
<tr><td>A</td><td>Growth Main Miles</td></tr>
<tr><td>B</td><td>Proactive and Reactive Main Replacement Miles</td></tr>
<tr><td>C</td><td>Reliability & Reinforcements Miles</td></tr>
<tr><td>D</td><td>CSC MRPL (includes Encroachments) Miles</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>E</td><td>Avg. Feet Per Day - Growth Main</td></tr>
<tr><td>F</td><td>Avg. Feet Per Day - Proactive and Reactive Main Replacement</td></tr>
</table>
</td>
<td>M&C Workplan</td>
</tr>
</table>

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

<table>
<tr>
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<td></td>
<td></td>
<td>
<table>
<tr><td>G</td><td>Avg. Feet Per Day - Reliability & Reinforcements</td></tr>
<tr><td>H</td><td>Avg. Feet Per Day - CSC MRPL (includes Encroachments)</td></tr>
<tr><td>J</td><td>Jurisdiction Average (Mains)</td></tr>
<tr><td>K</td><td>The number of gas in-house / contractors FTEs (based on average crew size during the Fiscal Year)</td></tr>
</table>

Variables E, F, G, and H are calculated as follows for in-house:
<table>
<tr><th>EFGH</th><th>Average Feet Per Day (In-House)</th></tr>
<tr><td></td><td>EFGH = (I*L)/P*Q*O</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>L</td><td>5280 (# of feet in a mile)</td></tr>
<tr><td>O</td><td>FTEs per Crew (varies by region)</td></tr>
<tr><td>P</td><td>Total Hours Worked</td></tr>
<tr><td>Q</td><td>8 hours daily</td></tr>
</table>

Variables E, F, G, and H are calculated as follows for contractors:
<table>
<tr><th>EFGH</th><th>Average Feet Per Day (Contractor)</th></tr>
<tr><td></td><td>EFGH = R*O</td></tr>
<tr><td>R</td><td>Average benchmark breakdown from Gas Construction (Fixed Annually)</td></tr>
<tr><td>O</td><td>FTEs per Crew (varies by region) (Fixed Annually)</td></tr>
</table>

An Excel file illustrating the calculation is included as Appendix 10a.
</td>
<td></td>
</tr>
<tr>
<td>Engineering and Back Office</td>
<td>Work Orders Processed/ Back Office FTEs (WOs/yr)</td>
<td>Measures the average number of Work Orders processed per year</td>
<td>The quotient of (i) Company total Work Orders in STORMS during the fiscal year <u>divided by</u> (ii) average number of Clerks active during the fiscal year.

"Work Orders" means work orders marked as closed in STORMS</td>
<td>STORMS Data</td>
</tr>
</table>

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

		in the back office by each engineering clerical employee	"Clerks" means those Company and Service Company employee and contractor roles responsible for processing Work Orders. The number of Clerks responsible for processing Work Orders utilized in calculating the KPI baseline is 26 comprising of: Administrative Clerk A (Niagara Mohawk employees – 6) Junior Clerk (Niagara Mohawk employee – 1) Contractors (Niagara Mohawk – 9, National Grid USA Service Company, Inc. – 10) For purposes of calculating actual KPI performance for each Rate Year as required by Section 7.8.4 of the JP and for FY23 under Section 7.8.3 of the JP, the Company will calculate the number of Clerks for the given FY based on the roles listed above for Niagara Mohawk and National Grid USA, Inc. Service Company.	R850 completion Clerical information (HRIS) and filtered for those responsible for processing work orders
Customer	Total Call Volume (Calls/year)	Measures the total volume of move and non-move telephone calls received by the Company's call centers from customers	The sum of (i) the total number of Move Calls received by the call centers from Company's customers during the fiscal year *plus* (ii) the total number of field-related Non-move Calls received by the call centers from Company's customers during the same measurement window. "Move Calls" means calls for service installation or discontinuance. "Non-move Calls" means calls to resolve a customer issue.	Move Call Benefits – call center query for fiscal year Non Move Call Benefits – call center query for fiscal year Move and Non-move Calls are included but not

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

				differentiated in the total Annual Call Volumes reported in the annual Service Quality Assurance Program report (discussed in Appendix 15) provided on a calendar year basis.
Customer / Key Measurement	Customer Experience (Effort) Rating	Measures the overall satisfaction of National Grid residential customers captured in the "Ease of Doing Business" survey, the results of which are mainly driven by number handoffs between Company departments and ease of issue resolutions	Average of scores for the "Ease of Doing Business" survey question to Company customers during the fiscal year, with the result expressed to two decimal places.	Ease of Doing Business Results from the Brand, Image & Relationship Survey

GBE Program Niagara Mohawk ("Company") KPI Descriptions and Calculations

Gas Safety & Compliance	Total non-compliance occurrences	Measures the total number of gas safety non-compliance occurrences	Total occurrences of violations of the Commission's gas safety regulations, as measured by the final results of Staff's annual records and field audits.	NMPC Records and Field Audit Results Cumulative Annual Non-Compliance Occurrences (as discussed in Joint Proposal Section 12.5) from Pipeline Safety & Compliance Team

Baseline:

values are assumed based on MA data

New York - NiMo (New Calculation using footage)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	34	miles	62	feet/day	103,133	4.5	21.16		
	Proactive and Reactive Main Replacement	45	miles	51	feet/day	167,619	4.5	28.29		
	Reliability & Reinforcements	2	miles	43	feet/day	10,669	4.5	1.50		
	CSC MRPL (includes Encroachements)	5	miles	47	feet/day	20,809	4.5	3.20		
	Total	**86**					**4.5**	**54.14**	**12.03**	**12.39**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	39	33	29	feet/day/crew	34
Proactive and Reactive Main Replacement	35	43	55	feet/day/crew	44
Proactive and Reactive Main Replacement	0	1	1	feet/day/crew	1
CSC MRPL (includes Encroachements)	6	4	5	feet/day/crew	5
CSC MRPL (includes Encroachements)	0	0	0	feet/day/crew	0
Reliability & Reinforcements	1	4	2	feet/day/crew	2
Total	**81**	**85**	**92**		86

NiMo Current Calculation

FY18

Growth Main	28	miles	62	feet/day	85,181	4.5	16.82		
Proactive and Reactive Main Replacement	51	miles	51	feet/day	189,116	4.5	30.72		
Reliability & Reinforcements	3	miles	43	feet/day	15,332	4.5	2.08		
CSC MRPL (includes Encroachements)	6	miles	47	feet/day	24,297	4.5	3.60		Feet per FTE
Total	**88**					**4.5**	**53.22**		**11.83** FY18

FY19

Growth Main	18	miles	62	feet/day	56,598	4.5	12.94		
Proactive and Reactive Main Replacement	50	miles	51	feet/day	186,109	4.5	35.00		
Reliability & Reinforcements	5	miles	43	feet/day	22,695	4.5	3.56		
CSC MRPL (includes Encroachements)	1	miles	47	feet/day	5,791	4.5	0.99		Feet per FTE
Total	**75**					**4.5**	**52.49**		**11.66** FY19

FY20

Growth Main	24	miles	62	feet/day	73,801	4.5	15.82		
Proactive and Reactive Main Replacement	46	miles	51	feet/day	170,796	4.5	30.11		
Reliability & Reinforcements	7	miles	43	feet/day	32,194	4.5	4.73		
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	12,467	4.5	2.00		Feet per FTE
Total	**80**				**289,258**	**4.5**	**52.67**		**11.70** FY20

FY21

Growth Main	17	miles	62	feet/day	50,770	4.5	12.44		
Proactive and Reactive Main Replacement	48	miles	51	feet/day	179,219	4.5	36.13		
Reliability & Reinforcements	4	miles	43	feet/day	16,139	4.5	2.71		
CSC MRPL (includes Encroachements)	2	miles	47	feet/day	6,860	4.5	1.26		Feet per FTE
Total	**70**				**252,988**	**4.5**	**52.54**		**11.68** FY21

FY22

Growth Main	17	miles	62	feet/day	53,328	4.5	13.96		
Proactive and Reactive Main Replacement	46	miles	51	feet/day	169,710	4.5	36.55		
Reliability & Reinforcements	1	miles	43	feet/day	3,381	4.5	0.61		
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	10,377	4.5	2.04		Feet per FTE
Total	**66**				**236,797**	**4.5**	**53.16**		**11.81** FY22

324

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 17	miles	(E) 169	feet/day	16,292	3.8	44.42		
Proactive and Reactive Main Replacement	(B) 46	miles	(F) 139	feet/day	51,789	3.8	95.76		
Reliability & Reinforcements	(C) 1	miles	(G) 154	feet/day	777	3.8	1.76		
CSC MRPL (includes Encroachements)	(D) 3	miles	(H) 125	feet/day	3,211	3.8	4.80		Feet per FTE
Total	**(I) 66**				**72,069**	**(K) 3.8**	**(J) 146.74**		**39.13** FY22

Baseline:

values are assumed based on MA data

New York - KEDNY (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	18	miles	64	feet/day	53,033	4.5	10.56		
	Proactive and Reactive Main Replacement	21	miles	54	feet/day	75,319	4.5	10.68		
	Reliability & Reinforcements	12	miles	43	feet/day	51,592	4.5	4.69		
	CSC MRPL (includes Encroachements)	57	miles	48	feet/day	225,465	4.5	25.52		
	Total	**108**					**4.5**	**51.44**	**11.43**	**11.77**

Data FY16 - FY17	Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
	Growth Main	18	16	20	feet/day/crew	18
	Proactive and Reactive Main Replacement	17	18	29	feet/day/crew	21
	Reliability & Reinforcements	14	10	11	feet/day/crew	12
	CSC MRPL (includes Encroachements)	38	67	67	feet/day/crew	57
	Total	**88**	**111**	**126**		**108**

(Original Baseline Calculation)

FY18										
	Growth Main	9	miles	64	feet/day	28,019	4.5	5.05		
	Proactive and Reactive Main Replacement	31	miles	54	feet/day	108,581	4.5	13.92		
	Reliability & Reinforcements	8	miles	43	feet/day	33,496	4.5	2.76		
	CSC MRPL (includes Encroachements)	72	miles	48	feet/day	282,578	4.5	28.93		Feet per FTE
	Total	**120**					**4.5**	**50.66**		**11.26** FY18

FY19										
	Growth Main	7	miles	64	feet/day	22,184	4.5	4.46		
	Proactive and Reactive Main Replacement	35	miles	54	feet/day	121,515	4.5	17.39		
	Reliability & Reinforcements	7	miles	43	feet/day	32,152	4.5	2.95		
	CSC MRPL (includes Encroachements)	58	miles	48	feet/day	228,147	4.5	26.06		Feet per FTE
	Total	**107**					**4.5**	**50.86**		**11.30** FY19

FY20										
	Growth Main	5	miles	64	feet/day	14,504	4.5	3.16		
	Proactive and Reactive Main Replacement	41	miles	54	feet/day	143,457	4.5	22.22		
	Reliability & Reinforcements	9	miles	43	feet/day	38,451	4.5	3.82		
	CSC MRPL (includes Encroachements)	45	miles	48	feet/day	175,955	4.5	21.76		Feet per FTE
	Total	**99**				**372,367**	**4.5**	**50.95**		**11.32** FY20

FY21										
	Growth Main	3	miles	64	feet/day	9,803	4.5	3.28		
	Proactive and Reactive Main Replacement	18	miles	54	feet/day	63,055	4.5	15.03		
	Reliability & Reinforcements	4	miles	43	feet/day	19,355	4.5	2.96		
	CSC MRPL (includes Encroachements)	39	miles	48	feet/day	152,651	4.5	29.05		Feet per FTE
	Total	**64**				**244,864**	**4.5**	**50.32**		**11.18** FY21

FY22										
	Growth Main	2	miles	64	feet/day	6,914	4.5	1.93		
	Proactive and Reactive Main Replacement	29	miles	54	feet/day	102,867	4.5	20.46		
	Reliability & Reinforcements	4	miles	43	feet/day	17,237	4.5	2.20		
	CSC MRPL (includes Encroachements)	42	miles	48	feet/day	164,038	4.5	26.05		Feet per FTE
	Total	**77**				**291,055**	**4.5**	**50.65**		**11.26** FY22

377

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22										
	Growth Main	(A) 2	miles	(E) 18	feet/day	27,816	5.0	0.53		
	Proactive and Reactive Main Replacement	(B) 29	miles	(F) 47	feet/day	132,369	5.0	17.67		
	Reliability & Reinforcements	(C) 4	miles	(G) 37	feet/day	22,242	5.0	1.89		
	CSC MRPL (includes Encroachements)	(D) 42	miles	(H) 50	feet/day	174,600	5.0	27.20		Feet per FTE
	Total	**(I) 77**				**357,027**	**(K) 5.0**	**(J) 47.30**		**9.46** FY22

Baseline:

values are assumed based on MA data

New York - KEDLI (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	54	miles	64	feet/day	158,945	4.5	20.31		
	Proactive and Reactive Main Replacement	97	miles	54	feet/day	343,137	4.5	31.21		
	Reliability & Reinforcements	13	miles	43	feet/day	56,730	4.5	3.31		
	CSC MRPL (includes Encroachements)	5	miles	48	feet/day	18,753	4.5	1.36		
	Total	**169**					**4.5**	**56.19**	**12.49**	**12.86**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	59	47	55	feet/day/crew	54
Proactive and Reactive Main Replacement	79	96	118	feet/day/crew	97
Reliability & Reinforcements	15	14	10	feet/day/crew	13
CSC MRPL (includes Encroachements)	4	3	7	feet/day/crew	5
Total	**156**	**160**	**190**		**169**

(Original Baseline Calculation)

FY18

Growth Main	51	miles	64	feet/day	150,787	4.5	17.76	
Proactive and Reactive Main Replacement	117	miles	54	feet/day	412,182	4.5	34.55	
Reliability & Reinforcements	13	miles	43	feet/day	58,074	4.5	3.12	
CSC MRPL (includes Encroachements)	2	miles	48	feet/day	7,561	4.5	0.51	
Total	**183**					**4.5**	**55.94**	

Feet per FTE **12.43** FY18

Rounded:

FY19

Growth Main	58	miles	64	feet/day	172,130	4.5	16.30	
Proactive and Reactive Main Replacement	160	miles	54	feet/day	564,589	4.5	38.07	
Reliability & Reinforcements	8	miles	43	feet/day	37,284	4.5	1.61	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	2,622	4.5	0.14	
Total	**227**					**4.5**	**56.13**	

Feet per FTE **12.47** FY19

FY20

Growth Main	31	miles	64	feet/day	91,444	4.5	9.63	
Proactive and Reactive Main Replacement	166	miles	54	feet/day	585,336	4.5	43.90	
Reliability & Reinforcements	7	miles	43	feet/day	31,355	4.5	1.51	
CSC MRPL (includes Encroachements)	0	miles	48	feet/day	1,229	4.5	0.07	
Total	**205**				**709,363**	**4.5**	**55.12**	

Feet per FTE **12.25** FY20

FY21

Growth Main	23	miles	64	feet/day	69,755	4.5	11.25	
Proactive and Reactive Main Replacement	103	miles	54	feet/day	363,501	4.5	41.75	
Reliability & Reinforcements	6	miles	43	feet/day	25,212	4.5	1.86	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	4,277	4.5	0.39	
Total	**134**				**462,745**	**4.5**	**55.25**	

Feet per FTE **12.28** FY21

FY22

Growth Main	32	miles	64	feet/day	94,586	4.5	11.54	
Proactive and Reactive Main Replacement	137	miles	54	feet/day	482,638	4.5	41.94	
Reliability & Reinforcements	7	miles	43	feet/day	28,992	4.5	1.62	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	3,951	4.5	0.27	
Total	**177**				**610,167**	**4.5**	**55.37**	

Feet per FTE **12.30** FY22

863

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 32	miles	(E) 105	feet/day	54,576	4.3	18.90	
Proactive and Reactive Main Replacement	(B) 137	miles	(F) 167	feet/day	147,038	4.3	130.00	
Reliability & Reinforcements	(C) 7	miles	(G) 148	feet/day	8,022	4.3	5.52	
CSC MRPL (includes Encroachements)	(D) 1	miles	(H) 26	feet/day	7,013	4.3	0.15	
Total	**(I) 177**				**216,649**	**(K) 4.3**	**(J) 154.56**	

Feet per FTE **36.37** FY22

KEDNY

Contractor	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	8,245	5.0	5.1	220.0	7.30	36.5
Proactive and Reactive Main Replacement	25	133,560	5.0	60.9	220.0	9.97	49.8
Reliability & Reinforcements	4	20,589	5.0	11.8	220.0	7.94	39.7
CSC MRPL (includes Encroachements)	34	178,734	5.0	65.8	220.0	12.35	61.8
Total	**65**	**341,128**	**5.0**	**143.6**	**220.0**	**10.80**	**53.99**

In-House	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	1	4,046	5.0	12.4	176.0	1.85	9.2
Proactive and Reactive Main Replacement	4	20,740	5.0	22.6	176.0	5.20	26.0
Reliability & Reinforcements	0	119	5.0	2.3	176.0	0.30	1.5
CSC MRPL (includes Encroachements)	8	41,123	5.0	44.5	176.0	5.25	26.3
Total	**13**	**66,028**	**5.0**	**81.8**	**176.0**	**4.59**	**22.93**

ALL	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	12,291	5.0	17.6	198.0	3.53	18
Proactive and Reactive Main Replacement	29	154,300	5.0	83.6	198.0	9.33	47
Reliability & Reinforcements	4	20,708	5.0	14.0	198.0	7.45	37
CSC MRPL (includes Encroachements)	42	219,857	5.0	110.2	198.0	10.07	50
Total	**(I) 77**	**407,156**	**(O) 5.0**	**(M) 225.4**	**(N) 198.0**		**(EFGH)**

KEDLI

Contractor	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	86,011	3.3	5.0	16.7	220	23.48	117.39
Proactive and Reactive Main Replacement	104	546,775	10.9	5.0	54.4	220	45.71	228.57
Reliability & Reinforcements	5	26,097	0.8	5.0	3.9	220	30.61	153.06
CSC MRPL (includes Encroachements)	1	5,296	0.9	5.0	4.4	220	5.50	27.51
Total	**126**	**664,179**	**15.9**	**5.0**	**79.3**	**220**	**38.08**	**190.42**

In-House	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	82,141	4.8	3.5	16.7	176	27.92	97.71
Proactive and Reactive Main Replacement	34	177,182	11.0	3.5	38.4	176	26.22	91.79
Reliability & Reinforcements	2	8,734	0.4	3.5	1.5	176	34.03	119.10
CSC MRPL (includes Encroachements)	-	-	0.2	3.5	0.6	176	-	-
Total	**51**	**268,057**	**16.3**	**3.5**	**57.1**	**176**	**26.65**	**93.28**

ALL	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	32	168,152	8.1	4.3	34.5	198.0	24.65	104.76
Proactive and Reactive Main Replacement	137	723,957	21.8	4.3	92.8	198.0	39.39	167.40
Reliability & Reinforcements	7	34,831	1.2	4.3	5.1	198.0	34.73	147.62
CSC MRPL (includes Encroachements)	1	5,296	1.0	4.3	4.4	198.0	6.04	25.68
Total	**(I) 177**	**932,236**	**32.2**	**(O) 4.3**	**(M) 136.8**	**(N) 198.0**		**(EFGH)**

NiMo

Contractor	FY22 (Miles)	FY22 (Feet)	Hours (Not Avail.)	Feet Per Hour (Not Avail.)	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	4	18,558	-	-	4.0	100.00	400.0
Replacement Main	29	153,259	-	-	4.0	50.00	200.0
DOT	2	8,055	-	-	4.0	50.00	200.0
Encroachments	-	-	-	-	4.0	-	-
Uprate	-	-	-	-	4.0	-	-
Total	**34**	**179,872**		**-**	**(O) 4.0**	**(R)**	**(EFGH)**

In-House	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	14	73,284	18,795	3.9	3.5	31.19	109.2
Replacement Main	17	87,164	48,004	1.8	3.5	14.53	50.8
DOT	1	5,349	5,048	1.1	3.5	8.48	29.7
Encroachments	0	888	5,519	0.2	3.5	1.29	4.5
Uprate	1	3,104	939	3.3	3.5	26.45	92.6
Total	**(I) 32**	**169,789**	**(P) 78,304**	**2.2**	**(O) 3.5**		**(EFGH)**

ALL	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	17	91,842	-	-	3.75	45.1	169.1
Replacement Main	46	240,423	-	-	3.75	37.1	139.3
DOT	3	13,404	-	-	3.75	33.4	125.2
Encroachments	0	888	-	-	3.75	1.3	4.8
Uprate	1	3,104	-	-	3.75	26.4	99.2
Total	**66**	**349,661**		**-**	**3.75**	**-**	

Attachment 3

KEDNY/KEDLI KPI Document

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

<table>
<tr><th>Measurement Area</th><th>KPI</th><th>Metric Description</th><th>Algorithm</th><th>Data Source</th></tr>
<tr><td>Work Management-Customer Meter Services ("CMS")</td><td>Avg. # of completed Jobs per CMS FTE (jobs/day)</td><td>Measures the average number of jobs completed per day by each working CMS FTE</td><td>The quotient of (i) the total number of Jobs completed by CMS FTEs during the fiscal year, divided by (ii) the total number of CMS FTEs during the fiscal year, with the result divided by the number of working days (240) and expressed to two decimal places.

CMS KPI: A/B/C
<table><tr><th>Variables</th><th>Description</th></tr><tr><td>A</td><td>Total CMS Jobs Completed including Collections</td></tr><tr><td>B</td><td>Total CMS FTE Count</td></tr><tr><td>C</td><td>Working Days in a Year</td></tr></table>"FTEs" includes Company employees and excludes contractors

"Jobs" means all Company CMS planned and collections jobs completed by CMS FTEs

"Jobs" utilized to calculate the KPI baseline and the actual KPI performance are listed below:

<u>KEDNY</u>

<u>CMS Planned Jobs</u>
Customer Driven
Construction – Capital
Gas Leak Investigation</td><td>Jobs Data and HRIS system</td></tr>
</table>

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

			Investigations Meter / Regulator Work Meter Change - Capital Meter Oriented Services Misc. - Capital Other Emergency Related to Meter Reading Surveillance Collections Jobs: • Completed Visits - Terminations and Collections **KEDLI** CMS Planned Jobs: Emergency Gas - O&M Turn-On Turn-Off – O&M Investigate and Read Meter – O&M Inspection-Asset – O&M Leak Surveillance, Recheck and Inspection – O&M Leak Monitoring – O&M Meter Asset Work – Capital Maintain Service – Capital Collections Jobs: Completed Visits – Terminations and Collections	
Work Management-Maintenance &	Avg. ft. of gas distribution main installed or	Measures the average feet of gas distribution	The formula to calculate the KPI is:	M&C Workplan

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

<table>
<tr>
<td>Construction ("M&C")</td>
<td>replaced per M&C and contractor FTE (ft/day)</td>
<td>main installed per day by each working gas M&C and contractor FTE</td>
<td>
M&C KPI $= \frac{J}{K}$ where
J = ((A*E/I) + (B*F/I) + (C*G/I) + (D*H/I))
K = Total number of gas in-house / contractor FTEs (based on average crew size during the fiscal year)

<table>
<tr><th>Variables</th><th>Description</th></tr>
<tr><td>A</td><td>Growth Main Miles</td></tr>
<tr><td>B</td><td>Proactive and Reactive Main Replacement Miles</td></tr>
<tr><td>C</td><td>Reliability & Reinforcements Miles</td></tr>
<tr><td>D</td><td>CSC MRPL (includes Encroachments) Miles</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>E</td><td>Avg. Feet Per Day - Growth Main</td></tr>
<tr><td>F</td><td>Avg. Feet Per Day - Proactive and Reactive Main Replacement</td></tr>
<tr><td>G</td><td>Avg. Feet Per Day - Reliability & Reinforcements</td></tr>
<tr><td>H</td><td>Avg. Feet Per Day - CSC MRPL (includes Encroachments)</td></tr>
<tr><td>J</td><td>Jurisdiction Average (Mains)</td></tr>
<tr><td>K</td><td>The number of gas in-house / contractors FTEs (based on average crew size during the Fiscal Year)</td></tr>
</table>
Variables E, F, G, and H are calculated as follows:
</td>
<td></td>
</tr>
</table>

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>
<table>
<tr><th>EFGH</th><th>Average Feet Per Day</th></tr>
<tr><td></td><td>EFGH = (I*L)/M/N*O</td></tr>
<tr><td>I</td><td>Total miles of gas Distribution Main installed in the Fiscal Year</td></tr>
<tr><td>L</td><td>5280 (# of feet in a mile)</td></tr>
<tr><td>M</td><td>The number of gas In-house / contractors FTEs (based on average crew size during the Fiscal Year)</td></tr>
<tr><td>N</td><td>Annual Productive Days (220 days for Contractors and 176 days for In-House)</td></tr>
<tr><td>O</td><td>FTEs per Crew (varies by region)</td></tr>
</table>
An Excel file illustrating the calculation is included as Appendix 10a.
</td>
<td></td>
</tr>
<tr>
<td>Engineering and Back Office</td>
<td>Work Orders Processed/ Back Office FTEs (WOs/yr)</td>
<td>Measures the average number of Work Orders processed per year in the back office by each engineering clerical employee</td>
<td>The quotient of (i) Company total Work Orders in Maximo during the fiscal year divided by (ii) average number of Clerks active during the fiscal year.

"Work Orders" means work orders marked as "Casbuilt" in Maximo.

"Clerks" means those Company and Service Company employee and contractor roles responsible for processing Work Orders.

For purposes of calculating actual KPI performance for each Rate Year as required by the JP and for FY24, the Company will calculate the number of Clerks for the given FY based on the</td>
<td>Maximo data Casbuilt work order completion

Clerical information (HRIS) and filtered for those responsible for</td>
</tr>
</table>

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

			number of clerks who are responsible for "Casbuilting" work order data during said year. The count of work orders "Casbuilt" during the FY will be only those that have been completed by the "Casbuilting" clerical staff for those work types included in the original KPI target calculation.	processing work orders
Customer	Total Call Volume (Calls/year)	Measures the total volume of move and non-move telephone calls received by the Company's call centers from customers	The sum of (i) the total number of Move Calls received by the call centers from Company's customers during the fiscal year *plus* (ii) the total number of field-related Non-move Calls received by the call centers from Company's customers during the same measurement window. "Move Calls" means calls for service installation or discontinuance. "Non-move Calls" means calls to resolve a customer issue.	Move Call Benefits – call center query for fiscal year Non Move Call Benefits – call center query for fiscal year Move and Non-move Calls are included but not differentiated in the total Annual Call Volumes reported in the annual Service Quality Assurance Program

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

				report (discussed in Section 9.0 of the Joint Proposal) provided on a calendar year basis.
Customer / Key Measurement	Customer Experience (Effort) Rating	Measures the overall satisfaction of National Grid residential customers captured in the "Ease of Doing Business" survey, the results of which are mainly driven by number handoffs between Company departments and ease of issue resolutions	Average of scores for the "Ease of Doing Business" survey question to Company customers during the fiscal year, with the result expressed to two decimal places.	Ease of Doing Business Results from the Brand, Image & Relationship Survey
Gas Safety & Compliance	Total non-compliance occurrences	Measures the total number of gas safety non-	Total occurrences of violations of the Commission's gas safety regulations, as measured by the final results of Staff's annual records and field audits.	KEDNY/KEDLI Records

GBE Program KEDNY and KEDLI ("Company") KPI Descriptions and Calculations

		compliance occurrences		and Field Audit Results Cumulative Annual Non-Compliance Occurrences (as discussed in Joint Proposal Section 10.5) from Pipeline Safety & Compliance Team

Baseline:

values are assumed based on MA data

New York - NiMo (New Calculation using footage)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	34	miles	62	feet/day	103,133	4.5	21.16		
	Proactive and Reactive Main Replacement	45	miles	51	feet/day	167,619	4.5	28.29		
	Reliability & Reinforcements	2	miles	43	feet/day	10,669	4.5	1.50		
	CSC MRPL (includes Encroachements)	5	miles	47	feet/day	20,809	4.5	3.20		
	Total	**86**					**4.5**	**54.14**	**12.03**	**12.39**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	39	33	29	feet/day/crew	34
Proactive and Reactive Main Replacement	35	43	55	feet/day/crew	44
Proactive and Reactive Main Replacement	0	1	1	feet/day/crew	1
CSC MRPL (includes Encroachements)	6	4	5	feet/day/crew	5
CSC MRPL (includes Encroachements)	0	0	0	feet/day/crew	0
Reliability & Reinforcements	1	4	2	feet/day/crew	2
Total	**81**	**85**	**92**		86

NiMo Current Calculation

FY18

Growth Main	28	miles	62	feet/day	85,181	4.5	16.82		
Proactive and Reactive Main Replacement	51	miles	51	feet/day	189,116	4.5	30.72		
Reliability & Reinforcements	3	miles	43	feet/day	15,332	4.5	2.08		
CSC MRPL (includes Encroachements)	6	miles	47	feet/day	24,297	4.5	3.60		
Total	**88**					**4.5**	**53.22**		

Feet per FTE **11.83** FY18

FY19

Growth Main	18	miles	62	feet/day	56,598	4.5	12.94	
Proactive and Reactive Main Replacement	50	miles	51	feet/day	186,109	4.5	35.00	
Reliability & Reinforcements	5	miles	43	feet/day	22,695	4.5	3.56	
CSC MRPL (includes Encroachements)	1	miles	47	feet/day	5,791	4.5	0.99	
Total	**75**					**4.5**	**52.49**	

Feet per FTE **11.66** FY19

FY20

Growth Main	24	miles	62	feet/day	73,801	4.5	15.82	
Proactive and Reactive Main Replacement	46	miles	51	feet/day	170,796	4.5	30.11	
Reliability & Reinforcements	7	miles	43	feet/day	32,194	4.5	4.73	
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	12,467	4.5	2.00	
Total	**80**				**289,258**	**4.5**	**52.67**	

Feet per FTE **11.70** FY20

FY21

Growth Main	17	miles	62	feet/day	50,770	4.5	12.44	
Proactive and Reactive Main Replacement	48	miles	51	feet/day	179,219	4.5	36.13	
Reliability & Reinforcements	4	miles	43	feet/day	16,139	4.5	2.71	
CSC MRPL (includes Encroachements)	2	miles	47	feet/day	6,860	4.5	1.26	
Total	**70**				**252,988**	**4.5**	**52.54**	

Feet per FTE **11.68** FY21

FY22

Growth Main	17	miles	62	feet/day	53,328	4.5	13.96	
Proactive and Reactive Main Replacement	46	miles	51	feet/day	169,710	4.5	36.55	
Reliability & Reinforcements	1	miles	43	feet/day	3,381	4.5	0.61	
CSC MRPL (includes Encroachements)	3	miles	47	feet/day	10,377	4.5	2.04	
Total	**66**				**236,797**	**4.5**	**53.16**	

Feet per FTE **11.81** FY22

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Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 17	miles	(E) 169	feet/day	16,292	3.8	44.42	
Proactive and Reactive Main Replacement	(B) 46	miles	(F) 139	feet/day	51,789	3.8	95.76	
Reliability & Reinforcements	(C) 1	miles	(G) 154	feet/day	777	3.8	1.76	
CSC MRPL (includes Encroachements)	(D) 3	miles	(H) 125	feet/day	3,211	3.8	4.80	
Total	**(I) 66**				**72,069**	**(K) 3.8**	**(J) 146.74**	

Feet per FTE **39.13** FY22

Baseline:

values are assumed based on MA data

New York - KEDNY (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	18	miles	64	feet/day	53,033	4.5	10.56		
	Proactive and Reactive Main Replacement	21	miles	54	feet/day	75,319	4.5	10.68		
	Reliability & Reinforcements	12	miles	43	feet/day	51,592	4.5	4.69		
	CSC MRPL (includes Encroachements)	57	miles	48	feet/day	225,465	4.5	25.52		
	Total	**108**					**4.5**	**51.44**	**11.43**	**11.77**

Data FY16 - FY17	Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
	Growth Main	18	16	20	feet/day/crew	18
	Proactive and Reactive Main Replacement	17	18	29	feet/day/crew	21
	Reliability & Reinforcements	14	10	11	feet/day/crew	12
	CSC MRPL (includes Encroachements)	38	67	67	feet/day/crew	57
	Total	**88**	**111**	**126**		**108**

(Original Baseline Calculation)

FY18	Growth Main	9	miles	64	feet/day	28,019	4.5	5.05		
	Proactive and Reactive Main Replacement	31	miles	54	feet/day	108,581	4.5	13.92		
	Reliability & Reinforcements	8	miles	43	feet/day	33,496	4.5	2.76		
	CSC MRPL (includes Encroachements)	72	miles	48	feet/day	282,578	4.5	28.93		Feet per FTE
	Total	**120**					**4.5**	**50.66**		**11.26** FY18
FY19	Growth Main	7	miles	64	feet/day	22,184	4.5	4.46		
	Proactive and Reactive Main Replacement	35	miles	54	feet/day	121,515	4.5	17.39		
	Reliability & Reinforcements	7	miles	43	feet/day	32,152	4.5	2.95		
	CSC MRPL (includes Encroachements)	58	miles	48	feet/day	228,147	4.5	26.06		Feet per FTE
	Total	**107**					**4.5**	**50.86**		**11.30** FY19
FY20	Growth Main	5	miles	64	feet/day	14,504	4.5	3.16		
	Proactive and Reactive Main Replacement	41	miles	54	feet/day	143,457	4.5	22.22		
	Reliability & Reinforcements	9	miles	43	feet/day	38,451	4.5	3.82		
	CSC MRPL (includes Encroachements)	45	miles	48	feet/day	175,955	4.5	21.76		Feet per FTE
	Total	**99**				**372,367**	**4.5**	**50.95**		**11.32** FY20
FY21	Growth Main	3	miles	64	feet/day	9,803	4.5	3.28		
	Proactive and Reactive Main Replacement	18	miles	54	feet/day	63,055	4.5	15.03		
	Reliability & Reinforcements	4	miles	43	feet/day	19,355	4.5	2.96		
	CSC MRPL (includes Encroachements)	39	miles	48	feet/day	152,651	4.5	29.05		Feet per FTE
	Total	**64**				**244,864**	**4.5**	**50.32**		**11.18** FY21
FY22	Growth Main	2	miles	64	feet/day	6,914	4.5	1.93		
	Proactive and Reactive Main Replacement	29	miles	54	feet/day	102,867	4.5	20.46		
	Reliability & Reinforcements	4	miles	43	feet/day	17,237	4.5	2.20		
	CSC MRPL (includes Encroachements)	42	miles	48	feet/day	164,038	4.5	26.05		Feet per FTE
	Total	**77**				**291,055**	**4.5**	**50.65**		**11.26** FY22

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Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22	Growth Main	(A) 2	miles	(E) 18	feet/day	27,816	5.0	0.53		
	Proactive and Reactive Main Replacement	(B) 29	miles	(F) 47	feet/day	132,369	5.0	17.67		
	Reliability & Reinforcements	(C) 4	miles	(G) 37	feet/day	22,242	5.0	1.89		
	CSC MRPL (includes Encroachements)	(D) 42	miles	(H) 50	feet/day	174,600	5.0	27.20		Feet per FTE
	Total	**(I) 77**				**357,027**	**(K) 5.0**	**(J) 47.30**		**9.46** FY22

Baseline:

values are assumed based on MA data

New York - KEDLI (Original Calculation Filed with PSC)	Program	3 Year Average FY16-FY18	Actuals Unit	Average Productivity/Unit	Productivity Unit	Average Yearly Productivity Hours	FTE/Unit	Jurisdiction Average (Mains)	2016 Baseline (Feet per FTE)	FY24 Target (Feet per FTE)
	Growth Main	54	miles	64	feet/day	158,945	4.5	20.31		
	Proactive and Reactive Main Replacement	97	miles	54	feet/day	343,137	4.5	31.21		
	Reliability & Reinforcements	13	miles	43	feet/day	56,730	4.5	3.31		
	CSC MRPL (includes Encroachements)	5	miles	48	feet/day	18,753	4.5	1.36		
	Total	**169**					**4.5**	**56.19**	**12.49**	**12.86**

Data FY16 - FY17

Program (aligned to other OpCos)	FY16 Work Unit Actuals	FY17 Work Unit Actuals	FY18 Work Unit Budget	Units	3 Year Average
Growth Main	59	47	55	feet/day/crew	54
Proactive and Reactive Main Replacement	79	96	118	feet/day/crew	97
Reliability & Reinforcements	15	14	10	feet/day/crew	13
CSC MRPL (includes Encroachements)	4	3	7	feet/day/crew	5
Total	**156**	**160**	**190**		**169**

(Original Baseline Calculation)

FY18

Growth Main	51	miles	64	feet/day	150,787	4.5	17.76	
Proactive and Reactive Main Replacement	117	miles	54	feet/day	412,182	4.5	34.55	
Reliability & Reinforcements	13	miles	43	feet/day	58,074	4.5	3.12	
CSC MRPL (includes Encroachements)	2	miles	48	feet/day	7,561	4.5	0.51	
Total	**183**					**4.5**	**55.94**	

Feet per FTE **12.43** FY18

Rounded:

FY19

Growth Main	58	miles	64	feet/day	172,130	4.5	16.30	
Proactive and Reactive Main Replacement	160	miles	54	feet/day	564,589	4.5	38.07	
Reliability & Reinforcements	8	miles	43	feet/day	37,284	4.5	1.61	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	2,622	4.5	0.14	
Total	**227**					**4.5**	**56.13**	

Feet per FTE **12.47** FY19

FY20

Growth Main	31	miles	64	feet/day	91,444	4.5	9.63	
Proactive and Reactive Main Replacement	166	miles	54	feet/day	585,336	4.5	43.90	
Reliability & Reinforcements	7	miles	43	feet/day	31,355	4.5	1.51	
CSC MRPL (includes Encroachements)	0	miles	48	feet/day	1,229	4.5	0.07	
Total	**205**				**709,363**	**4.5**	**55.12**	

Feet per FTE **12.25** FY20

FY21

Growth Main	23	miles	64	feet/day	69,755	4.5	11.25	
Proactive and Reactive Main Replacement	103	miles	54	feet/day	363,501	4.5	41.75	
Reliability & Reinforcements	6	miles	43	feet/day	25,212	4.5	1.86	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	4,277	4.5	0.39	
Total	**134**				**462,745**	**4.5**	**55.25**	

Feet per FTE **12.28** FY21

FY22

Growth Main	32	miles	64	feet/day	94,586	4.5	11.54	
Proactive and Reactive Main Replacement	137	miles	54	feet/day	482,638	4.5	41.94	
Reliability & Reinforcements	7	miles	43	feet/day	28,992	4.5	1.62	
CSC MRPL (includes Encroachements)	1	miles	48	feet/day	3,951	4.5	0.27	
Total	**177**				**610,167**	**4.5**	**55.37**	

Feet per FTE **12.30** FY22

863

Adjusting the Average Productivity Per Unit and FTE Per Unit

FY22

Growth Main	(A) 32	miles	(E) 105	feet/day	54,576	4.3	18.90	
Proactive and Reactive Main Replacement	(B) 137	miles	(F) 167	feet/day	147,038	4.3	130.00	
Reliability & Reinforcements	(C) 7	miles	(G) 148	feet/day	8,022	4.3	5.52	
CSC MRPL (includes Encroachements)	(D) 1	miles	(H) 26	feet/day	7,013	4.3	0.15	
Total	**(I) 177**				**216,649**	**(K) 4.3**	**(J) 154.56**	

Feet per FTE **36.37** FY22

KEDNY

Contractor	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	8,245	5.0	5.1	220.0	7.30	36.5
Proactive and Reactive Main Replacement	25	133,560	5.0	60.9	220.0	9.97	49.8
Reliability & Reinforcements	4	20,589	5.0	11.8	220.0	7.94	39.7
CSC MRPL (includes Encroachements)	34	178,734	5.0	65.8	220.0	12.35	61.8
Total	**65**	**341,128**	**5.0**	**143.6**	**220.0**	**10.80**	**53.99**

In-House	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	1	4,046	5.0	12.4	176.0	1.85	9.2
Proactive and Reactive Main Replacement	4	20,740	5.0	22.6	176.0	5.20	26.0
Reliability & Reinforcements	0	119	5.0	2.3	176.0	0.30	1.5
CSC MRPL (includes Encroachements)	8	41,123	5.0	44.5	176.0	5.25	26.3
Total	**13**	**66,028**	**5.0**	**81.8**	**176.0**	**4.59**	**22.93**

ALL	FY22 (Miles)	FY22 (Feet)	FTEs per Crew	# of FTEs	Annual Productive Days	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
Growth Main	2	12,291	5.0	17.6	198.0	3.53	18
Proactive and Reactive Main Replacement	29	154,300	5.0	83.6	198.0	9.33	47
Reliability & Reinforcements	4	20,708	5.0	14.0	198.0	7.45	37
CSC MRPL (includes Encroachements)	42	219,857	5.0	110.2	198.0	10.07	50
Total	**(I) 77**	**407,156**	**(O) 5.0**	**(M) 225.4**	**(N) 198.0**		**(EFGH)**

KEDLI

Contractor	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	86,011	3.3	5.0	16.7	220	23.48	117.39
Proactive and Reactive Main Replacement	104	546,775	10.9	5.0	54.4	220	45.71	228.57
Reliability & Reinforcements	5	26,097	0.8	5.0	3.9	220	30.61	153.06
CSC MRPL (includes Encroachements)	1	5,296	0.9	5.0	4.4	220	5.50	27.51
Total	**126**	**664,179**	**15.9**	**5.0**	**79.3**	**220**	**38.08**	**190.42**

In-House	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	16	82,141	4.8	3.5	16.7	176	27.92	97.71
Proactive and Reactive Main Replacement	34	177,182	11.0	3.5	38.4	176	26.22	91.79
Reliability & Reinforcements	2	8,734	0.4	3.5	1.5	176	34.03	119.10
CSC MRPL (includes Encroachements)	-	-	0.2	3.5	0.6	176	-	-
Total	**51**	**268,057**	**16.3**	**3.5**	**57.1**	**176**	**26.65**	**93.28**

ALL	FY22 (Miles)	FY22 (Feet)	Avg # of Crews	FTEs per Crew	# of FTEs	Annual Productive Days	Main Install per FTE/per day (feet)	Main Install per crew / per day (feet)
Growth Main	32	168,152	8.1	4.3	34.5	198.0	24.65	104.76
Proactive and Reactive Main Replacement	137	723,957	21.8	4.3	92.8	198.0	39.39	167.40
Reliability & Reinforcements	7	34,831	1.2	4.3	5.1	198.0	34.73	147.62
CSC MRPL (includes Encroachements)	1	5,296	1.0	4.3	4.4	198.0	6.04	25.68
Total	**(I) 177**	**932,236**	**32.2**	**(O) 4.3**	**(M) 136.8**	**(N) 198.0**		**(EFGH)**

NiMo

Contractor	FY22 (Miles)	FY22 (Feet)	Hours (Not Avail.)	Feet Per Hour (Not Avail.)	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	4	18,558	-	-	4.0	100.00	400.0
Replacement Main	29	153,259	-	-	4.0	50.00	200.0
DOT	2	8,055	-	-	4.0	50.00	200.0
Encroachments	-	-	-	-	4.0	-	-
Uprate	-	-	-	-	4.0	-	-
Total	**34**	**179,872**		**-**	**(O) 4.0**	**(R)**	**(EFGH)**

In-House	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	14	73,284	18,795	3.9	3.5	31.19	109.2
Replacement Main	17	87,164	48,004	1.8	3.5	14.53	50.8
DOT	1	5,349	5,048	1.1	3.5	8.48	29.7
Encroachments	0	888	5,519	0.2	3.5	1.29	4.5
Uprate	1	3,104	939	3.3	3.5	26.45	92.6
Total	**(I) 32**	**169,789**	**(P) 78,304**	**2.2**	**(O) 3.5**		**(EFGH)**

ALL	FY22 (Miles)	FY22 (Feet)	Hours	Feet Per Hour	Average Crew Size	Main Installation per FTE / per day (feet)	Main Installation per crew / per day (feet)
New Main	17	91,842	-	-	3.75	45.1	169.1
Replacement Main	46	240,423	-	-	3.75	37.1	139.3
DOT	3	13,404	-	-	3.75	33.4	125.2
Encroachments	0	888	-	-	3.75	1.3	4.8
Uprate	1	3,104	-	-	3.75	26.4	99.2
Total	**66**	**349,661**		**-**	**3.75**	**-**	